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Summary

Reference Number	NIHR304617
Lead Applicant	Dr Weiyu Ye
Research Title	Predicting long-term outcomes in people with psoriasis who achieve disease control on systemic therapy: a mixed methods study
Plain English Summary	**Background** Psoriasis is a common skin condition that affects 1 in 50 people in the UK. It causes red scaly patches on the skin and there is currently no known cure. For people with severe psoriasis, tablet or injection treatments are often used to help them get it under control. Newer injectable treatments called 'biologics' work better than older treatments and have helped many more people get their psoriasis under control initially. Unfortunately, not everyone can keep it under control in the long-term, even if they stay on treatment. Currently, we cannot predict if a person will keep their psoriasis under control when they stay on a particular treatment. For people with psoriasis, not knowing how long a particular treatment will work for can be extremely worrying. It also means that everyone on treatment must attend regular hospital check-ups, even if they do not actually need them. Limited NHS resources are therefore being used inefficiently, and people who need appointments may struggle to get one. **Aim** To create a tool to identify people who are more or less likely to keep their psoriasis under control while receiving a tablet or injection treatment. This could help to personalise their long-term care and improve their treatment experience. **Methods** First, I will review existing research to see how common it is for people to keep their psoriasis under control on tablet or injection treatments. I will also look for individual factors, such as age and weight, which could make this more or less likely. Using two large datasets from the UK and Europe, I will then build computer models to predict how likely it is for someone to keep their psoriasis under control. This prediction will be based on routinely collected clinical data, such as age, ethnicity and smoking history. I will also look at whether including genetic data (i.e. information stored in DNA) improves how well the models perform. Finally, I will host separate group discussions with healthcare professionals (including doctors and nurses) and people with psoriasis. They will provide input on which models would be best suited for use as a tool. They will also help to make the tool more user-friendly. After creating the tool, I will invite healthcare professionals to test it, and further improvements will be made based on their feedback. **Patient and Public Involvement** The problem this project aims to address has been identified as a research priority by people with psoriasis, their families and friends, and the healthcare professionals who treat them. The design of the project incorporates further feedback from a meeting I had with people with psoriasis. For example, they emphasised that involving patients and healthcare professionals from the start is essential for developing a useful tool. Patients and the public will continue to be involved during this project through 6-monthly meetings. I will invite them to help plan the group discussions and analyse the conversations after the sessions. They will also ensure the research is inclusive and relevant to diverse communities. **Impact** This work will help people better understand how likely they are to keep their psoriasis under control while receiving a tablet or injection treatment. It will also allow healthcare professionals to provide better care matched to individual needs. This will improve the patient experience and allow healthcare resources to be used more efficiently.

	Dissemination I will share my findings by presenting at conferences and by publishing in scientific journals. I will also work with patients and the public to develop easily understood summaries which will be circulated widely through charities, newsletters and social media.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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1. Application Summary Information

Host organisation (which will administer any award)
King's College London

Partner organisation (if applicable)	

Application Type

Strategic Theme	Health data science

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Predicting long-term outcomes in people with psoriasis who achieve disease control on systemic therapy: a mixed methods study

Research Type	Epidemiological, Qualitative study, Evidence Synthesis

Proposed start date if grant awarded	01/10/2024

2. Applicant CV

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2022	10/2022	Specialty Certificate Examination in Dermatology	Specialty Certificate Examination in Dermatology	United Kingdom	Royal Colleges of Physicians of the United Kingdom and British Association of Dermatologists	Pass	Royal Colleges of Physicians of the United Kingdom and British Association of Dermatologists
10/2020	01/2024	Postgraduate Diploma (PGDip) Health Research	Health Research	United Kingdom	University of Oxford	Distinction	Department for Continuing Education
01/2019	12/2020	MRCP(UK)	Membership of the Royal Colleges of Physicians of the United Kingdom	United Kingdom	Royal Colleges of Physicians of the United Kingdom	Pass	Royal Colleges of Physicians of the United Kingdom
10/2011	07/2017	MBBChir	Medicine	United Kingdom	University of Cambridge	Distinction	Medicine
10/2011	07/2014	BA	Preclinical Medical Sciences	United Kingdom	University of Cambridge	First Class	Medicine

Present and previous positions

(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
08/2022	-	Specialty Registrar in Dermatology	HEE Thames Valley	0
08/2019	07/2022	NIHR Academic Clinical Fellow in Dermatology	HEE Thames Valley	25
08/2018	07/2019	Academic Foundation Year 2 Doctor	HEE Thames Valley	33
07/2017	07/2018	Academic Foundation Year 1 Doctor	HEE Thames Valley	6

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
Cancer Research UK Predoctoral Bursary	Lead Applicant & Principle Investigator	25000	Cancer Research UK	12	01/05/2021	01/05/2022

Publication Record

(include the list of authors, article title, journal name, and volume)

This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Theodoulou, A., Chepkin, SC., Ye, W., Fanshawe, TR., Bullen, C., Hartmann-Boyce, J., Livingstone-Banks, J., Hajizadeh, A. and Lindson, N.. (2023) ‘Different doses, durations and modes of delivery of nicotine replacement therapy for smoking cessation.’, *The Cochrane database of systematic reviews*, 6.

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Fairfax, B., Taylor, C., Watson, R., Tong, O., Ye, W., Nassiri, I., Gilchrist, J., Aires, AVdl., Sharma, P., Koturan, S., Cooper, R., Woodcock, V., Jungkurth, E., Shine, B., Coupe, N., Payne, M., Naranbhai, V., Groha, S., Emery, P., Mankia, K., Freedman, M., Choueiri, T. and Gusev, A.. (2022) ‘Genetic variation at IL7 provides mechanistic insights into toxicity to immune checkpoint blockade’,
Ye, W., Kim, M., Fairfax, BP., Coupe, N., Payne, MJ. and Matin, RN.. (2021) ‘Phenotyping of cutaneous toxicities in patients with metastatic malignant melanoma treated with immune checkpoint blockade therapy at a UK tertiary care centre.’, *Clinical and experimental dermatology*, 47(2).
Ye, W., Hackett, S., Vandevelde, C., Twigg, S., Helliwell, PS. and Coates, LC.. (2021) ‘Measuring Physical Function in Psoriatic Arthritis: Comparing the Multidimensional Health Assessment Questionnaire to the Health Assessment Questionnaire-Disability Index.’, *The Journal of rheumatology*, 48(11).
Ye, W., Olsson-Brown, A., Watson, RA., Cheung, VTF., Morgan, RD., Nassiri, I., Cooper, R., Taylor, CA., Akbani, U., Brain, O., Matin, RN., Coupe, N., Middleton, MR., Coles, M., Sacco, JJ., Payne, MJ. and Fairfax, BP.. (2021) ‘Checkpoint-blocker-induced autoimmunity is associated with favourable outcome in metastatic melanoma and distinct T-cell expression profiles.’, *British Journal of Cancer*, 124(10).
Ye, W., Hackett, S., Vandevelde, C., Twigg, S., Helliwell, PS. and Coates, LC.. (2020) ‘Comparing the Visual Analog Scale and the Numerical Rating Scale in Patient-reported Outcomes in Psoriatic Arthritis.’, *The Journal of rheumatology*, 48(6).
Eagle, DM., Schepisi, C., Chugh, S., Desai, S., Han, SYS., Huang, T., Lee, JJ., Sobala, C., Ye, W., Milton, AL. and Robbins, TW.. (2020) ‘Dissociable dopaminergic and pavlovian influences in goal-trackers and sign-trackers on a model of compulsive checking in OCD.’, *Psychopharmacology*, 237(12).
Ye, W. and Coates, LC.. (2019) ‘Should Methotrexate Have Any Place in the Treatment of Psoriatic Arthritis?’, *Rheumatic diseases clinics of North America*, 45(3).
Lindson, N., Chepkin, SC., Ye, W., Fanshawe, TR., Bullen, C. and Hartmann-Boyce, J.. (2019) ‘Different doses, durations and modes of delivery of nicotine replacement therapy for smoking cessation.’, *The Cochrane database of systematic reviews*, 4.
Ye, W., Tucker, LJ. and Coates, LC.. (2018) ‘Tapering and Discontinuation of Biologics in Patients with Psoriatic Arthritis with Low Disease Activity.’, *Drugs*, 78(16).
Tucker, LJ., Ye, W. and Coates, LC.. (2018) ‘Novel Concepts in Psoriatic Arthritis Management: Can We Treat to Target?’, *Current rheumatology reports*, 20(11).
Hartmann-Boyce, J., Chepkin, SC., Ye, W., Bullen, C. and Lancaster, T.. (2018) ‘Nicotine replacement therapy versus control for smoking cessation.’, *The Cochrane database of systematic reviews*, 5.
Stober, C., Ye, W., Guruparan, T., Htut, E., Clunie, G. and Jadon, D.. (2018) ‘Prevalence and predictors of tumour necrosis factor inhibitor persistence in psoriatic arthritis.’, *Rheumatology (Oxford, England)*, 57(1).
Ye, W., Fifield, MC. and Östör, AJ.. (2017) ‘Refractory Rheumatoid Arthritis and Associated Interstitial Lung Disease: Could Abatacept be the Answer?’, *Journal of clinical rheumatology : practical reports on rheumatic & musculoskeletal diseases*, 23(2).
Georgiou, C., Raja, M., Ye, W., Grosu, L., Jaffar, H., Neale, E. and Mahran, M.. (2015) ‘Re: Pelvic organ prolapse and incontinence 15-23 years after first delivery: a cross-sectional study.’, *BJOG : an international journal of obstetrics and gynaecology*, 122(12).
Ye, W., Fifield, MC., Mayhew, A., Nasser, S. and Östör, A.. (2015) ‘Successful tocilizumab desensitization in an adult with juvenile idiopathic arthritis.’, *Scandinavian journal of rheumatology*, 45(1).
Ye, W. and Norris, JM.. (2015) ‘Deep brain stimulation for Parkinson disease in elderly individuals.’, *JAMA neurology*, 72(3).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/12/2023	UK Dermatology Clinical Trials Network (UKDCTN)	UKDCTN Neil Cox SpR Fellowship Award	Fellowship Award
01/10/2023	European Academy of Dermatology & Venereology	Michael Hornstein Memorial Scholarship	Scholarship winner
01/07/2023	World Congress of Dermatology	Conference Scholarship	Scholarship winner
01/06/2023	European Alliance of Associations for Rheumatology (EULAR)	Fully funded Travel Bursary to attend the Annual European Congress of Rheumatology	Prize winner
01/11/2019	The British Psoriatic Arthritis Consortium (BritPACT)	Brit-PACT Runner Up Abstract Prize	Prize winner
01/11/2018	The British Psoriatic Arthritis Consortium (BritPACT)	Brit-PACT Runner Up Abstract Prize	Prize winner
01/07/2017	University of Cambridge	Distinction in Final M.B. Examinations	Prize winner
01/07/2017	Jesus College, University of Cambridge	Waring Prize for Part III of the Final M.B. Examinations	Prize winner
01/07/2017	Jesus College, University of Cambridge	Academic Exhibition for Examination Performance	Prize winner
01/05/2017	University of Cambridge	Jim Knott Essay Prize in Hospital Palliative Care	Prize winner
01/07/2016	Jesus College, University of Cambridge	Academic Exhibition for Examination Performance	Prize winner
01/07/2016	University of Cambridge	Distinction in Pathology Examinations	Prize winner
01/07/2015	University of Cambridge	William Harvey Studentship for Examination Performance	Prize winner
01/05/2015	University of Cambridge	Bettie Knott Essay Prize in Palliative Care	Prize winner
01/07/2014	Jesus College, University of Cambridge	Academic Scholarship for Examination Performance	Prize winner
01/07/2013	Jesus College, University of Cambridge	Academic Scholarship for Examination Performance	Prize winner
01/07/2013	The Evelyn Trust	Evelyn Trust Studentship for Summer Research	Award holder

ORCID iD 0000-0002-8983-0355

3. Applicant Research Background

Professional Background	Medically Qualified

Specialty (if Medically Qualified)
Dermatology

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	7552607

Please describe your research career to date

I aspire to be an academic dermatologist, and have successfully followed a clinical academic training path to date. The NIHR Doctoral Research Fellowship (DRF) is a natural next step on my journey, enabling me to hone in on my particular interest in developing personalised management strategies for inflammatory skin disease. This will prepare me well for subsequent postdoctoral training through an NIHR Clinical Lectureship and for my eventual goal of spearheading cutting-edge research in this area as an independent Clinician Scientist.

Career Background

I am a Dermatology Registrar in the Thames Valley Deanery. I studied Medicine at the University of Cambridge, completing an intercalated degree in Neuroscience in 2014 (First Class Honours, ranked 2nd in cohort), and graduating with Distinction in 2017. During this period, I won 3 university and 5 college prizes for academic excellence. Following graduation, I completed a highly competitive Academic Foundation Programme (AFP) and an NIHR Academic Clinical Fellowship (ACF) in Dermatology. I have a proven track-record of successfully balancing research with clinical training, having authored 14 peer-reviewed publications with 8 as first author (*h*-index 9). I am well-equipped for this DRF due to my extensive research experience and strong organisational skills.

Research Experience

During my AFP, I extracted data and assessed the risk of bias for two systematic reviews for the Cochrane Tobacco Addiction Group, following a 4-day Training Programme organised by Cochrane UK. I also worked with Professor Laura Coates, an academic rheumatologist, to design and run a multicentre study comparing different patient-reported outcome measures in psoriatic arthritis. During this, I completed Good Clinical Practice training, secured ethical approval, wrote the study documents, coordinated patient recruitment and analysed the data. I presented my findings at the 2019 British Psoriatic Arthritis Consortium, winning the runner-up abstract award, with two subsequent first author publications in *The Journal of Rheumatology*.

In parallel, I worked on a multicentre cohort study examining the association between immune-related toxicity and survival in metastatic melanoma patients treated with checkpoint immunotherapy. Supervised by Professor Benjamin Fairfax, an academic oncologist, I led the project from conception to completion and became proficient in using R for descriptive statistics and survival analysis. My work resulted in a joint first author publication in the *British Journal of Cancer*. Separately, I characterised the cutaneous toxicities in this cohort, which led to an oral presentation at the 2020 Annual Meeting of the British Association of Dermatologists (BAD) and a first author publication in *Clinical and Experimental Dermatology*.

As part of my subsequent ACF in Dermatology, I completed a PGDip in Health Research with Distinction at the University of Oxford. This provided in-depth training on study design, statistics, diagnostic and screening studies, systematic reviews, and statistical computing using R and Stata. For my 9-month research rotation, I secured £25,000 from Cancer Research UK. This funded single-cell RNA sequencing experiments to investigate gene expression changes in melanoma patients who develop immune-related toxicities following checkpoint immunotherapy. In addition to in-house training on peripheral mononuclear cell extraction, immune cell subset isolation and single-cell sequencing, I attended courses on Linux for Genomics, RNA-seq data analysis and R to complement my self-directed learning. My work contributed to a co-author publication in *Nature Medicine* and I won two sought-after travel scholarships to present my research (European Congress of Rheumatology, World Congress of Dermatology).

Since completing my ACF, I have immersed myself in full-time dermatology training while actively contributing to research. I am a peer reviewer for the journals *Skin Health and Disease* and the *British Journal of Dermatology* and have recently been awarded a prestigious part-time fellowship from the UK Dermatology Clinical Trials Network (receiving the joint-highest score), where I will receive further training in patient-focused clinical research.

My diverse research experiences to date have helped me realise that my true passion lies in clinical research with a direct impact on patient care. Being halfway through specialist training, I have experienced many different subspecialties and identified inflammatory skin disease as an area combining my academic and clinical interests. I became a trainee representative for the British Association of Dermatologists Biologics and Immunomodulators Register (BADBIR), a national registry for people with psoriasis on systemic therapy. In addition, I have also recently joined the BAD Biologic Therapy for Psoriasis Guideline Development Group as a trainee representative. These experiences have highlighted critical unmet needs in psoriasis research, and have provided inspiration for this DRF proposal, which has been subsequently refined through discussions with supervisors and PPI collaborators.

Teaching Experience

I have taught medical students in classroom, ward and outpatient settings. I also designed and implemented a well-received junior doctor-led teaching programme for medical students during my AFP. Additionally, I have completed a Developing Teaching and Learning for Medical Educators course, and have supervised junior colleagues for multiple audits and quality improvement projects.

Leadership and Management Experience

During my AFP, I sat on the Steering Committee of a national conference for junior doctors and medical students to showcase their research. Within my department, I chaired weekly clinicopathological correlation meetings over a 6-month period, and I am currently responsible for organising monthly regional complex cases meetings.

Public Engagement Experience

I recently organised a PPI advisory event, collecting valuable feedback on my project proposal. I hope to utilise this fellowship to build further experience in public engagement and involvement, supported by the extensive expertise within our group.

Fellowship Impact

This fellowship provides the perfect platform to advance my career as an academic dermatologist, through an ambitious project which could transform long-term management for people with psoriasis on systemic therapy. At this pivotal stage in my training, taking an extended period away from clinical work will allow me to concentrate fully on my growth as a researcher. I am confident that I will be able to maximise this opportunity, due to my robust

training plan, my expert supervisory team, and the dynamic research environment at KCL. Completing this fellowship will position me well for a Clinical Lectureship, and bring me one step closer to my goal of becoming an impactful Clinician Scientist.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding, or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

4. Plain English Summary of Research

Background

Psoriasis is a common skin condition that affects 1 in 50 people in the UK. It causes red scaly patches on the skin and there is currently no known cure. For people with severe psoriasis, tablet or injection treatments are often used to help them get it under control.

Newer injectable treatments called 'biologics' work better than older treatments and have helped many more people get their psoriasis under control initially. Unfortunately, not everyone can keep it under control in the long-term, even if they stay on treatment.

Currently, we cannot predict if a person will keep their psoriasis under control when they stay on a particular treatment. For people with psoriasis, not knowing how long a particular treatment will work for can be extremely worrying. It also means that everyone on treatment must attend regular hospital check-ups, even if they do not actually need them. Limited NHS resources are therefore being used inefficiently, and people who need appointments may struggle to get one.

Aim

To create a tool to identify people who are more or less likely to keep their psoriasis under control while receiving a tablet or injection treatment. This could help to personalise their long-term care and improve their treatment experience.

Methods

First, I will review existing research to see how common it is for people to keep their psoriasis under control on tablet or injection treatments. I will also look for individual factors, such as age and weight, which could make this more or less likely.

Using two large datasets from the UK and Europe, I will then build computer models to predict how likely it is for someone to keep their psoriasis under control. This prediction will be based on routinely collected clinical data, such as age, ethnicity and smoking history. I will also look at whether including genetic data (i.e. information stored in DNA) improves how well the models perform.

Finally, I will host separate group discussions with healthcare professionals (including doctors and nurses) and people with psoriasis. They will provide input on which models would be best suited for use as a tool. They will also help to make the tool more user-friendly. After creating the tool, I will invite healthcare professionals to test it, and further improvements will be made based on their feedback.

Patient and Public Involvement

The problem this project aims to address has been identified as a research priority by people with psoriasis, their families and friends, and the healthcare professionals who treat them. The design of the project incorporates further feedback from a meeting I had with people with psoriasis. For example, they emphasised that involving patients and healthcare professionals from the start is essential for developing a useful tool. Patients and the public will continue to be involved during this project through 6-monthly meetings. I will invite them to help plan the group discussions and analyse the conversations after the sessions. They will also ensure the research is

inclusive and relevant to diverse communities.

Impact

This work will help people better understand how likely they are to keep their psoriasis under control while receiving a tablet or injection treatment. It will also allow healthcare professionals to provide better care matched to individual needs. This will improve the patient experience and allow healthcare resources to be used more efficiently.

Dissemination

I will share my findings by presenting at conferences and by publishing in scientific journals. I will also work with patients and the public to develop easily understood summaries which will be circulated widely through charities, newsletters and social media.

5. Scientific Abstract

Background

Psoriasis is a common, incurable inflammatory skin disease associated with high psychosocial morbidity and economic impact. Systemic therapy is often required for individuals with moderate-to-severe psoriasis to achieve disease control (i.e. clear or almost clear skin).

Systemic therapy encompasses conventional systemic immunosuppressive therapy, as well as a growing list of newer, highly effective biologics. While the effectiveness of biologics has led to a rapidly expanding cohort who achieve disease control, there still exists no means of predicting whether this will persist longer-term at the individual level. Uncertainty over the longevity of disease control not only causes significant anxiety for people with psoriasis, but also results in the continuation of a one-size-fits-all healthcare model of regular follow-ups regardless of clinical need. This leads to poorer treatment experiences, inefficient resource allocation, and a worsening supply and demand mismatch in the NHS.

Aims and Objectives

This project aims to develop a risk stratification tool to predict the likelihood of sustaining psoriasis control on systemic therapy. To do this, I will:

- Evaluate published evidence on the proportion of individuals sustaining disease control over time and the factors associated with this.
- Model the likelihood of sustaining disease control on systemic therapy using real-world datasets.
- Optimise the predictive models for use as a risk stratification tool, with patient and clinician input.

Methods and Timelines

Phase 1 (Months 0-9): Conduct a systematic review to determine the proportion of individuals sustaining psoriasis control on systemic therapy over time, and the clinical and genetic factors associated with this.

Phase 2 (Months 7-24): Develop and validate predictive models of sustained disease control on systemic therapy using two large-scale real-world datasets from the UK and Europe to ensure generalisability. I will also explore whether including genetic data improves model performance.

Phase 3 (Months 19-36): Refine the predictive models into a risk stratification tool. Patient and clinician focus groups will determine clinically acceptable levels of model performance, influence the definition of risk categories, and guide user interface design. The tool will be beta tested via a clinician workshop.

PPI feedback has been integral to developing this proposal. PPI collaborators will be closely involved throughout, including assisting with recruitment strategies, co-facilitating focus groups, and helping with qualitative analysis. They will also ensure the research is inclusive and relevant to diverse communities.

Impact

Successful delivery of the risk stratification tool could reduce patient anxiety, empower individuals to address

modifiable risk factors, and personalise long-term management strategies. For example, individuals likely to sustain disease control will benefit from increased autonomy and flexibility and may be suited to remote monitoring or patient-initiated follow-up pathways. Higher-risk individuals may be appropriate for adjunct therapies with closer monitoring, facilitating earlier intervention for flares to reduce disease impact.

Dissemination

My findings will be disseminated widely through high-impact peer-reviewed publications and presentations at national and international conferences. I will co-develop Plain English Summaries and infographics with my PPI collaborators, which will be shared via UK and European psoriasis patient associations, newsletters, and social media platforms to maximise public engagement.

6. Detailed Research Plan

1. Problem

Chronic plaque psoriasis ('psoriasis') is an immune-mediated inflammatory disease (IMID) affecting 2% of the UK population[1]. It presents as red scaly skin plaques and carries high psychosocial morbidity, with a quality-of-life impact comparable to cancer and cardiovascular disease[2]. Psoriasis also generates economic drag, with associated absenteeism and presenteeism costing c. £1.07 billion/year[3].

As psoriasis is a lifelong condition with no known cure, the goal of treatment is to achieve disease control (i.e. clear/almost clear skin[4]) and sustain this long-term. Individuals with moderate-to-severe disease are typically treated with conventional systemic therapy, escalating to biologics if needed[5]. Currently, 12 biologics are NICE-approved for psoriasis[5]. Although expensive (£3,000-10,000/patient/year[6]), they are highly effective. Individuals treated with adalimumab (TNF inhibitor) are twice as likely to achieve disease control compared to methotrexate, a first-line conventional systemic[7]. Newer biologics targeting IL-23 are even more effective, with 84% achieving disease control at 4 months compared to 60% with adalimumab[8].

Due to the success of biologics, more individuals with psoriasis are **achieving** disease control than ever before. However, the extent to which disease control can be **sustained** long-term varies, with no reliable way of predicting this currently. This uncertainty can cause significant anxiety[9] and leads to a one-size-fits-all healthcare model of regular follow-ups regardless of clinical need. This results in poorer treatment experiences, inefficient resource allocation, and compounds the supply and demand mismatch in the NHS, making it harder for those in need to access timely care[6,10].

This project aims to develop a risk stratification tool to predict the likelihood of **sustaining** psoriasis control on systemic therapy. Aligning with the NIHR strategic theme of Health Data Science, it will help personalise long-term management strategies. For example:

- Individuals likely to sustain disease control will benefit from increased autonomy and flexibility, and may be suitable for remote monitoring or patient-initiated follow-up (PIFU) pathways with less disruption to daily life. This will also save costs and release system capacity[10].
- Higher-risk individuals may be appropriate for adjunct therapies such as methotrexate, which may improve biologic persistence through reducing anti-drug antibody development[11], or phototherapy, which works synergistically with systemic treatments[12]. Closer monitoring will also facilitate earlier intervention for flares, thereby reducing disease impact.

Furthermore, this work may provide insights applicable to other IMIDs treated with systemic therapies (e.g. eczema, arthritis).

2. Importance

2.1. Stakeholders have highlighted improved risk stratification as a key research priority

This project seeks to improve stratification of individuals who have already achieved disease control based on their likelihood of sustaining it long-term. This aligns with 3 of the top 10 priorities in the 2018 James Lind Alliance Priority Setting Partnership involving patients, carers and clinicians[13]:

- 'What factors predict how well psoriasis responds to a treatment?'
- 'Why do psoriasis treatments stop working well?'

- 'What is the best way to treat sudden flare ups of psoriasis?'

People with active psoriasis often feel stigmatised and less confident due to its visual nature[14]. A global survey of >8,000 reported that clear/almost clear skin empowers individuals to undertake activities including swimming and wearing more varied clothing[14], highlighting the importance of sustaining disease control on wellbeing.

Participants from a PPI meeting I conducted were also strongly supportive of this research. One individual stated, 'Even though my skin has cleared on treatment, I am constantly worried about whether this will last and would like more certainty.' This underscores the potential for risk stratification to reduce anxiety.

2.2. Risk stratification tools can personalise the patient pathway, save costs and inform future research

Risk stratification is commonly used in routine care, as exemplified in the management of cancer[15] and cardiovascular disease[16]. The QRISK score is one of the best-known examples of a risk stratification tool and is NICE-recommended to guide statin initiation decisions[16]. Use of QRISK2 is associated with a higher proportion of statin initiations being consistent with NICE guidelines (85% vs 44% when not used[17]) and leads to substantial cost savings (c. £918 million within 2 years[18]).

Risk stratification tools have recently been developed for IMIDs, for example to guide blood monitoring frequency during long-term methotrexate[19] and thiopurine[20] treatment. Despite psoriasis being one of the most common IMIDs, no tools currently exist to predict the likelihood of sustaining disease control on systemic therapy. My project will address this gap, with potential to:

- Boost patient/clinician satisfaction by reducing uncertainty.
- Empower individuals to address modifiable risk factors.
- Personalise long-term management strategies, aligning with the NHS Long Term Plan[21].
- Inform participant selection for future trials, e.g. exploring dose minimisation strategies for cohorts likely to sustain disease control.

3. Literature

3.1. How is disease control defined in psoriasis?

The NICE psoriasis guidelines define disease control as a Physician's Global Assessment (PGA) of clear/almost clear skin[5]. This correlates to a Psoriasis Area and Severity Index (PASI) ≤2 or a PASI 90 response (i.e. 90% improvement from baseline)[4], where PASI is the gold standard disease severity measure in psoriasis[5].

Achievement of disease control is typically evaluated 2-6 months post-initiation of systemic therapy[22]. Whether this is **sustained** long-term (i.e. ≥6 months) is a separate, important issue, which my project will focus on. Loss of disease control (i.e. relapse) is defined as a ≥3 increase in PASI, with a minimum PASI of 5[23].

3.2. How feasible is sustained disease control on systemic therapy?

The Cochrane living systematic review comparing different systemic therapies only focused on randomised controlled trial (RCT) data ≤6 months post-initiation[22]. However, open-label extensions (OLE) to RCTs have shed light on the proportion sustaining longer-term disease control. After starting ixekizumab (IL-17 inhibitor), 75% and 73% of individuals exhibited disease control at 3 and 13 months[24]. In individuals who achieve disease control on risankizumab (IL-23 inhibitor) at 7 months post-initiation, 87% and 81% maintained this at 12 and 24 months[25].

Comparisons of longer-term RCT data are hindered by heterogeneity of reporting timepoints and trial design. Additionally, the demographics and complexity of trial populations differ from those seen in clinical practice[26]. To date, only one small-scale North American real-world registry study (n=687) has been published, indicating that a lower proportion of individuals sustain disease control on biologics outside of trial settings (73%/60%/50% at 6/12/18 months respectively)[27].

Although real-world drug persistence studies do not directly address the proportion sustaining disease control, they provide invaluable insight into longer-term effectiveness and safety. Data from the British Association of Dermatologists Biologics and Immunomodulators Register (BADBIR) demonstrates that 46% of individuals remain on methotrexate at 12 months[28], considerably lower than for those treated with biologics (75%-88% depending on the drug[29]). This highlights that drug choice may impact the likelihood of sustaining disease control.

Despite multiple studies looking at the proportion of individuals sustaining long-term psoriasis control, their clinical utility remains limited due to there being no comprehensive summary synthesising the key findings. My project will address this through a systematic review.

3.3. What clinical factors are associated with sustained disease control?

The rapidly growing cohort achieving initial psoriasis control is a recent phenomenon. As such, researching clinical factors associated with sustained disease control has only become a priority of late, with few studies to date **addressing this directly**. These include:

- A North American real-world registry study (n=687)[27], which identified skin itch (visual analogue scale, VAS ≥30/100) or skin pain (VAS ≥45/100) to be associated with increased relapse risk on biologics (HR=1.47; 95% CI 1.06-2.03). Conversely, being biologic-naïve at baseline was associated with reduced risk (HR=0.78; 95% CI 0.62-0.98).
- An analysis of guselkumab (IL-23 inhibitor) trial data (n=494)[30], which found diabetes (OR=0.23; 95% CI 0.05-0.98) and higher baseline disease impact (OR=0.94; 95% CI 0.89-0.99) to be associated with lower odds of sustaining disease control. In contrast, higher week 4 guselkumab plasma concentrations were associated with increased odds (OR=1.43, 95% CI 1.07-1.90).

However, both studies are limited by small sample sizes, wide confidence intervals and a lack of validation in independent datasets.

Further insight can be gleaned from studies investigating **different, but related, outcomes in psoriasis**. A meta-analysis of factors predicting biologic persistence[31] identified female sex (HR=1.22; 95% CI 1.07-1.38) and obesity (HR=1.21, 95% CI 1.10-1.32) to be associated with treatment discontinuation, while concomitant psoriatic arthritis was associated with a decreased likelihood of this (HR=0.83, 95% CI 0.80-0.86). This corroborates results from a BADBIR study[32] (n=3,079), where female sex (OR=0.78, 95% CI 0.66-0.93) and increased weight (OR=0.99, 95% CI 0.98-0.99) were linked to reduced odds of achieving PASI 90 at 12 months, alongside previous smoking (OR=0.76, 95% CI 0.63-0.93). Conversely, white ethnicity (OR=1.81, 95% CI 1.36-2.41) and concurrent psoriatic arthritis (OR=1.22, 95% CI 1.01-1.47) were associated with increased odds. A global self-report cross-sectional study[33] (n=4,043) led by our group found non-adherence to be associated with worsening psoriasis (OR=2.90, 95% CI 2.31-3.63), with an increased prevalence of depression/anxiety in the non-adherent cohort (42.8% vs 32.4% in the adherent cohort). Data from our NIHR-funded self-report platform[34] (n=555) suggest that patients

perceive stress, low mood and infections to be major reasons for worsening psoriasis.

Additionally, factors associated with **sustained disease control across other IMIDs** (which share many of the same systemic therapies) include smoking and joint inflammation in rheumatoid arthritis[35], shorter disease duration and high baseline CRP in psoriatic arthritis[36], and early treatment response in lupus[37]. Further research is required to establish whether these factors are also relevant in psoriasis.

Collectively, the literature suggests that risk stratification is possible for my target cohort. By incorporating both the direct and indirect research above, and applying their findings to large real-world datasets, I aim to build a clinically relevant risk stratification tool for routine care.

3.4. What are the genetic factors associated with sustained disease control?

There is a strong genetic component to psoriasis development, with previous research highlighting HLA-C*06:02 as a major risk factor for psoriasis susceptibility and severity[38]. HLA-C*06:02 also influences initial response to biologics, with a BADBIR study demonstrating HLA-C*06:02-negative patients to be significantly more likely to respond to adalimumab than ustekinumab (OR=2.95, 95% CI 1.93-4.51 at 6 months)[38]. Additionally, a recent scoping review by our group highlighted HLA-C*06:02 as the most extensively studied genetic factor associated with systemic treatment response in psoriasis[39], although further validation is needed prior to its routine use in isolation. Given that initial treatment response is a pre-requisite for longer-term disease control, it is plausible that HLA-C*06:02 status may predict the likelihood of sustaining disease control.

Recent work from our group also demonstrates that a psoriasis susceptibility polygenic risk score (PRS) based on 62 non-HLA single nucleotide polymorphisms associates with severe disease in the UK Biobank (OR=1.23, 95% CI 1.14-1.32), where 878 individuals with severe psoriasis were compared with 5,901 individuals with non-severe psoriasis[40]. While evidence from other diseases (e.g. schizophrenia[41]) indicates that PRS can be associated with treatment response, whether this is true for psoriasis remains unexplored.

Although routine genetic testing is not yet commonplace in the psoriasis population, this could soon change given the rapid expansion of genomic profiling technologies. My project will leverage recent progress in this field to determine whether combining genetic and clinical data can improve model performance in predicting the likelihood of sustained disease control.

3.5. Why is this research needed now?

Biologics have led to a paradigm shift in psoriasis treatment, with their effectiveness rapidly expanding the cohort achieving disease control. However, ongoing uncertainty over individual likelihoods of sustaining disease control has prevented their full benefits from being realised in real-world settings. Anxiety over potential relapse remains an issue, contributing to the persistence of legacy one-size-fits-all healthcare models designed for an era reliant on less effective treatments requiring more intensive monitoring.

My proposed research could deliver a step change in longer-term psoriasis management, improving treatment satisfaction and service efficiency. With the fast-growing adoption of targeted therapies for other IMIDs[42], my findings may also have broader impact beyond psoriasis.

4. Aims and objectives

My overarching aim is to develop a risk stratification tool to help tailor management for individuals with psoriasis who have achieved disease control on systemic therapy, thereby improving long-term outcomes.

To do this, I will:

- **Conduct a systematic review to evaluate published evidence on the proportion of individuals sustaining disease control over time and the factors associated with this (Phase 1).**
- **Model the likelihood of sustaining disease control on systemic therapy using large-scale real-world datasets (Phase 2).**
- **Optimise the predictive models for use as a risk stratification tool, with patient and clinician input (Phase 3).**

5. Project Plan

5.1. Phase 1: Systematic review (0-9 months)

Aim

Evaluate published evidence on the proportion of individuals who sustain psoriasis control on systemic therapy over time, and the factors associated with this.

Research questions

- 5.1a – What proportion of individuals sustain psoriasis control on systemic therapy over time?
- 5.1b – What clinical/genetic factors are associated with sustained psoriasis control on systemic therapy?

Design

I will conduct a systematic review in accordance with PRISMA guidelines[43]. A comprehensive search for primary studies will be performed in Medline, EMBASE, Web of Science and the Cochrane Skin Group Specialised Register. The protocol will be registered on PROSPERO.

I will leverage our group's collaboration with the Cochrane Skin Group[44], through which I will access a wealth of resources which could contribute to this review. These include:

- Completed data extraction for 167 RCTs with 12-month efficacy data, as part of a previous project evaluating study design heterogeneity[45].
- Ongoing data extraction from RCTs and observational studies for a systematic review of serious adverse events, which I will be supporting.

Eligible studies will meet the following criteria:

- **Study design** – RCTs (including OLE) and observational studies.
- **Population** – adults ≥18 years with psoriasis.

- **Intervention** – continued treatment with methotrexate or a biologic (TNF inhibitor, IL-12/IL-23 inhibitor, IL-17 inhibitor, IL-23 inhibitor) at least until the primary outcome is measured.
- **Comparator** – any psoriasis treatment, placebo or no comparator.
- **Primary outcome** – proportion with sustained disease control (i.e. PGA clear/almost clear, PASI ≤2, or a PASI 90 response) at 6 months after it is first achieved. In my PPI meeting, participants emphasised the importance of longer-term control (i.e. ≥6 months).
- **Secondary outcomes** – proportion with sustained disease control at later timepoints, effect sizes for clinical/genetic factors associated with sustained disease control, and proportion with serious adverse events.

Quality assessment

Risk of bias will be assessed using the ROBINS-I tool[46] for cohort studies/OLE and the ROB2 tool[47] for RCTs.

Data analysis

Data from extracted studies will be described narratively in terms of design, sample size and range of variables considered. To address Research Question (RQ) 5.1a, meta-regression models will be used to estimate the proportion sustaining disease control on systemic therapy over time. This will account for the likely variation in reporting at different follow-up time points. Depending on risk of bias and data heterogeneity, I will decide whether to progress to a meta-analysis of effect sizes for RQ5.1b. Where this is not possible, vote counting by effect size will be used to narratively synthesise the data[48].

5.2. Phase 2: Model development (7-24 months)

Aim

Develop predictive models of sustained disease control on systemic therapy using real-world data.

Research questions

- 5.2a – What proportion of individuals sustain disease control over time in real-world settings?
- 5.2b – How well can we predict the likelihood of sustained disease control on systemic therapy?

Datasets

In preparation for this fellowship, I have secured access to two large-scale real-world datasets.

BADBIR is a UK and Ireland prospective observational cohort study of people with psoriasis who have commenced systemic therapy within the previous 6 months[49]. Participants are recruited from >160 hospitals, with >18,000 people registered and >110,000 follow-ups to date. Comprehensive sociodemographic, clinical, treatment and disease severity data are collected at 6-monthly intervals for 3 years, then annually thereafter. Within BADBIR, >6,000 individuals have been genetically profiled via the Biomarkers and Stratification To Optimise outcomes in Psoriasis study (BSTOP), led by my supervisor Professor Smith[50]. The detailed phenotyping makes the BADBIR data ideal as a **discovery and internal validation dataset** for exploring RQ5.2a-b.

The HIPPOCRATES Prospective Observational Study (HPOS) is a cohort study across 12 European countries, part of a wider Innovative Medicines Initiative-funded EU consortium addressing unmet needs in psoriatic

disease[51]. It focuses on adults with self-reported psoriasis, without a pre-existing diagnosis of psoriatic arthritis, and is led by my supervisor, Professor Coates. Since opening in summer 2023, >1,300 people have enrolled, with a target of 25,000 by 2025. Participants self-enter sociodemographic, clinical, treatment and disease severity data every 6 months for 3 years. The self-reported data, overlap with the BADBIR data fields, and large sample size across multiple countries, make HPOS ideally suited for **externally validating** BADBIR findings.

Inclusion criteria

Adults ≥18 years with psoriasis who achieve disease control (i.e. PGA of clear/almost clear, PASI/self-assessed PASI ≤2 or PASI 90 response) post-initiation of systemic therapy, with ≥1 subsequent follow-up.

Where multiple measures are available (e.g. PGA and PASI), individuals will be included if all measures meet the inclusion threshold to ensure the cohort only includes individuals who have achieved disease control.

Preliminary filtering has identified c. 9,500 individuals from the BADBIR dataset with PASI ≤2, which is feasible for predictive model development (see sample size considerations).

As HPOS is still in the early stages of recruitment, estimating the eventual number meeting the inclusion criteria is challenging. To mitigate the risk of insufficient numbers by the time of my analysis, I will assess recruitment regularly throughout my first 6 months. If progress is insufficient, I will apply for access to PsoBest[52], a German national psoriasis registry. This has been used to validate a previous BADBIR study[53], with similar assessments performed at baseline and follow-up. Access is available through the BIOMAP consortium[54], a European collaboration coordinated by Professor Smith.

Data analysis

<u>RQ5.2a.</u>

Baseline characteristics of included participants will be summarised descriptively. Kaplan-Meier survival curves will show the proportion sustaining disease control over time. Observations will be censored at loss to follow-up or at 3 years post-registration, whichever is sooner. Using BADBIR data only, I will repeat this analysis stratifying by drug class (methotrexate, TNF inhibitor, IL-12/IL-23 inhibitor, IL-17 inhibitor, IL-23 inhibitor).

Both unadjusted and adjusted estimates will be reported. Unadjusted estimates from the Kaplan-Meier analysis will enable comparison with previous studies[27,30], and adjusted estimates will evaluate the impact of key sociodemographic variables (age/sex/ethnicity/location) on the unadjusted estimates. Adjusted estimates will be based on discrete-time cloglog survival curves, to determine the proportion sustaining disease control at 6-monthly intervals for up to 3 years. To account for repeated follow-ups within individuals, a random effect will be used.

Missing baseline clinical data will be accounted for by using multiple imputation methods, with estimates compared to a complete case sensitivity analysis. Missing outcome data due to missed follow-ups will be handled implicitly, under the assumption that data is missing at random. Additional sensitivity analyses will evaluate potential missing-not-at-random drop out mechanisms.

<u>RQ5.2b.</u>

Potential variables associated with sustained disease control will be selected based on Phase 1 results, expert opinion (leveraging my supervisors' clinical experience/networks), and PPI input, accounting for the extent of

missing data. Clinical variables will include sociodemographic, treatment, disease severity/impact and comorbidities (including mental health).

To account for differences in the variables available within the BADBIR and HPOS datasets, three separate approaches will be used:

- **Approach A will use clinical variables common to both datasets**. This will result in a simplified predictive model which is both internally and externally validated. This has potential to be used across different healthcare and resource settings, as only widely available variables will be included.

- **Approach B will use clinical variables in the more comprehensive BADBIR dataset**. This will result in a more complex predictive model which will be internally validated. This has the advantage of evaluating a broader range of variables compared to Approach A.

- **Approach C will focus on the subset of individuals in BADBIR with clinical and genetic data**, to explore whether addition of genetic data improves clinical predictive model performance.

This study will be reported according to the TRIPOD guidelines[55].

Approach A

Using clinical variables common to both BADBIR and HPOS, a penalised mixed-effects multivariable logistic regression model will be fit to the BADBIR dataset to predict the likelihood of sustained disease control for the next 6 months, controlling for length of time on treatment. A random effect will be included as individuals will contribute repeated measures of disease control at 6-monthly intervals. Missing data will be dealt with as in RQ5.2a.

Penalisation will use elastic net and lasso approaches to select the minimum subset of variables. I will also explore whether additional machine learning approaches (e.g. support vector machine, random forest, extreme gradient boosting) will improve model performance further. Methods including nested cross validation will be used for internal validation and to prevent overfitting.

The HPOS dataset will be used for external validation. Model performance will be assessed using measures of discrimination (C-statistic) and calibration (calibration slope, calibration-in-the-large).

Approach B

Using all clinical variables within the BADBIR dataset, I will fit, internally validate, and assess model performance using the methods outlined above. This analysis will also be repeated, stratifying by drug class, to evaluate model stability across different treatments.

Approach C

After filtering the BADBIR dataset to only include individuals with genetic data available, I will fit a penalised mixed-effects multivariable logistic regression model as outlined in Approach B. I will then incorporate data on HLA-C*06:02 and the polygenic risk score (see *Section 3.4*) individually to determine whether this improves model performance. This analysis will be repeated, stratifying by drug class.

Sample size considerations

Using methods described by Riley et al. 2019[56], the expected sample of c. 9,500 individuals from BADBIR is

sufficient for predictive model development and meets necessary criteria to minimise overfitting in a penalised logistic model. Using 80% of the cohort (c. 7,600) for model training and the rest for validation, the maximum number of potential predictors is 90 using Equation 11[56]; assuming an expected shrinkage factor of 0.90 and a conservative anticipated Cox-Snell R^2 of 0.10. A shrinkage factor of 0.90 is selected based on recommended practice[56], where the goal is small optimism in predictor effect estimates. Most predictive models in clinical practice use significantly fewer inputs[19,20,32], demonstrating the sufficiency of the BADBIR dataset. We calculated a maximum Cox-Snell R^2 of 0.74, based on a sample size of 7,600 and an outcome proportion of 60%[27].

5.3. Phase 3: Developing the risk stratification tool (19-36 months)

Aim

Refine Phase 2 predictive models into a risk stratification tool, utilising a co-design approach involving patients and clinicians to enhance acceptability for end-users.

Research questions

- 5.3a – Which models are clinically acceptable for development into a risk stratification tool?
- 5.3b – What is the best way to stratify a continuous risk spectrum into discrete risk groups?
- 5.3c – What barriers and enablers affect the use of a risk stratification tool, and how acceptable is it to use such a tool to guide psoriasis management?

Design

Phase 2 will produce a range of models designed to predict sustained disease control on systemic therapy, differing in their complexity, accuracy, and included variables. In Phase 3, I will use focus groups to assess stakeholder views on these models to inform subsequent tool development. The practicality of using the tool will be evaluated in a beta testing workshop. All participants will be reimbursed[57].

Focus groups

People with psoriasis and multidisciplinary healthcare professionals (dermatologists/pharmacists/nurses) will attend separate online focus groups for open discussion on RQ5.3a-c. These will be 1.5 hours each, consist of 6-8 participants, and be co-facilitated by a trained member of my Patient and Lay Advisory Group. A topic guide developed with PPI input will be used, with detailed probing followed by general questioning to achieve convergence to group norms[58].

Stakeholders will be shown the Phase 2 models, and their perspectives will inform:

- Acceptable levels of model performance for clinical use.
- The feasibility and acceptability of including specific variables (e.g. genetic data).
- How individuals with different risk profiles are categorised (e.g. number of groups/cutoff points).
- The tool's user interface and output presentation, with reference to examples provided.
- Implications for clinical practice and future postdoctoral work.

To enhance end-user generalisability, I will conduct ≥3 focus groups for healthcare professionals recruited from across the UK. The groups will be divided based on background and experience to minimise power imbalance. I will also conduct ≥3 focus groups for people with psoriasis recruited from the Psoriasis Association and dermatology clinics. I will leverage our skin research inclusion toolkit (manuscript in preparation) to ensure

representation from underserved groups, including lower socioeconomic classes, ethnic minorities and those with disabilities.

All discussions will be audio-recorded and transcribed verbatim to enable inductive thematic analysis using NVivo software. A systematic approach will be used to organise collected data into relevant codes, which will be grouped into potential themes using an iterative process[59]. I will involve two PPI collaborators in the data analysis to minimise bias.

These focus groups will establish stakeholder perspectives on required parameters for a clinically acceptable risk stratification tool. These perspectives are critical for maximising the clinical utility of risk stratification in practice and will provide generalisable insights beyond this project.

Beta testing

Informed by focus group discussions, I will refine the predictive models into a risk stratification tool, taking the form of an online interactive dashboard (built using R Shiny). This will be trialled in a 2-hour beta testing workshop involving 20 multidisciplinary healthcare professionals. This sample size is pragmatic, based on recruitment feasibility, and will provide sufficient data to determine tool usability for future work. The primary outcome will be the proportion who successfully input data to obtain an output, and the secondary outcome will be the data input time.

A structured questionnaire will collate feedback on user experience and potential solutions to any issues. To assess acceptability, participants will complete a separate questionnaire addressing the seven component constructs of the theoretical framework of acceptability: affective attitude, burden, ethicality, intervention coherence, opportunity cost, perceived effectiveness and self-efficacy[60].

The tool will be further improved based on the beta testing outcomes, with repeat testing sessions organised as required.

6. Dissemination/Outputs/Impact

6.1. Dissemination

Patients/Public

I will work with PPI collaborators to produce Plain English Summaries and infographics, which will be shared through the BADBIR newsletter and international consortia with well-established patient networks (HPOS/BIOMAP). Through our group's collaboration with the Psoriasis Association, my findings will also be disseminated via webinars and social media.

Clinicians/Researchers

My findings will be presented at national and international conferences, published in high-impact journals, and shared via the St John's DermAcademy platform. I will also use my supervisors' networks to reach other disciplines (e.g. rheumatology) where this work may be relevant.

Policymakers

My results will inform UK guideline updates via my publications and supervisors. Professor Smith chaired the NICE psoriasis guidelines committee. She also leads the British Association of Dermatologists systemic therapy guidelines committee where Professor Coates and Dr Mahil are both members.

6.2. Outputs and impact

Better risk stratification could yield significant benefits across stakeholder groups.

Patients

Reduce anxiety and empower individuals to change modifiable risk factors, improving satisfaction and long-term outcomes.

Clinicians

Facilitate earlier intervention for patients at increased risk of relapse, and provide more autonomy and flexibility for those at lower risk.

Researchers

Inform participant selection for future trials, e.g. those exploring dose minimisation strategies for cohorts likely to sustain disease control.

Commissioners

Contribute to more efficient service design, e.g. by identifying those most suitable for remote monitoring or PIFU pathways, so resources can be prioritised for individuals with the highest need.

7. Project management

7.1. Fellowship oversight

I will lead this project with support from my supervisors who will provide academic, clinical and methodological oversight. We will meet regularly to ensure my research and training progresses according to the Gantt chart.

My progress will also be monitored at King's College London via 6-monthly meetings with an independent Progression Committee.

7.2. Patient and Lay Advisory Group

I will meet this group (n=6) 6-monthly to discuss PPI strategy, gain feedback on progress, and plan results dissemination and implementation.

8. Ethics

8.1. Ethical approval

Ethical approval is in place for BADBIR (REC 07/MRE08/9), HPOS (23/PR/0418) and BSTOP (11/H0802-7). I will seek REC/HRA approval for Phase 3 during the fellowship.

8.2. Data protection and confidentiality

All data will be stored in compliance with the Data Protection Act 1998 and GDPR. Access will be restricted to the research team.

9. Success criteria and risks

9.1. Success criteria

Delivery of research to time and target (see *Section 4*).

Publications:

- Systematic review
- Predictive models of sustained disease control
- Co-design of risk stratification tool with patient and clinician input to predict the likelihood of sustaining disease control

Presentations to clinical, academic, lay and industry audiences.

Completion of my Training and Development Plan.

Development of a competitive NIHR Clinical Lectureship application.

9.2. Risks

Insufficient number of individuals meeting inclusion criteria in HPOS

In the unlikely event that insufficient numbers in HPOS meet the inclusion criteria by the time of my analysis, I will apply for access to PsoBest[52] as a contingency external validation dataset (see *Section 5.2 – Inclusion criteria*).

Failure to recruit sufficient participants for qualitative work

To encourage participation, virtual attendance (with a phone-in option) will be offered at convenient times.

Recruiting 24 individuals with psoriasis is feasible, based on PPI feedback, the experience of my supervisors, and our group's established links with the Psoriasis Association.

Recruiting 44 healthcare professionals may be more challenging. Allowing the same individuals to take part in the focus groups and beta testing reduces the number required to 24. I will leverage my supervisors' national networks, including their links to the UK Dermatology Clinical Trials Network (>1,000 healthcare professionals), to maximise reach.

Model performance falls below clinically acceptable thresholds

As outlined in *Section 3*, existing evidence supports the feasibility of risk stratification in my target cohort. With multiple predictive models being developed utilising a range of methods, the likelihood of at least one being acceptable is increased.

In the unlikely event that none are acceptable, the focus groups will still offer invaluable insight into patient and clinician perspectives on the required parameters for a clinically acceptable risk stratification tool. These perspectives are generalisable beyond this project and key to informing the direction of future research to maximise the clinical utility of risk stratification in practice.

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I originally developed this proposal with the findings of the 2018 James Lind Alliance Priority Setting Partnership[13] involving patients, carers and clinicians in mind. This highlighted research into factors affecting sustained disease control in psoriasis as a key unmet need, aligning with 3 of the top 10 priorities identified.

My proposal has since been refined further based on feedback from a PPI advisory group meeting I organised. Participants were recruited from local networks and the Psoriasis Association, with diversity in terms of age, ethnicity, geographical location and disease duration. They also had varied experiences of different topical and systemic therapies for psoriasis.

All participants strongly supported my research question and felt that it was relevant. In particular, they emphasised that:

- A better understanding of factors affecting sustained psoriasis control would reduce uncertainty for people on systemic therapy.
- Identification of modifiable risk factors may empower individuals to make appropriate lifestyle changes.
- Personalised follow-up strategies co-designed by patients and clinicians are preferable to a blanket approach.

Participants were enthusiastic about the idea of a risk stratification tool providing individualised estimates of the likelihood of sustained disease control on systemic therapy. The group highlighted the following key points, which have shaped my proposal:

- Disease control maintained over the longer term, i.e. ≥6 months, is especially important. Hence, Phases 1 and 2 focus on sustaining disease control for ≥6 months as a key outcome.
- Not everyone will be comfortable with genetic data being used. However, the group agreed that there is value in exploring whether genetic data can improve model performance. Consequently, Phase 2 will compare models including and excluding genetic data.
- To maximise the tool's clinical utility, early involvement of patients and clinicians is essential. This influenced Phase 3 of my proposal, in which stakeholder engagement is central to model refinement, user interface design and beta testing.

Following this event, 4 individuals expressed an interest in continued involvement in the project as part of a Patient and Lay Advisory Group. They have also provided helpful feedback for the proposal's Plain English Summary.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

PPI is integral to the success of my research, particularly given its potential to provide more autonomy and flexibility for individuals likely to sustain disease control long-term. To this end, I will establish a Patient and Lay Advisory Group of 6 individuals, including people with psoriasis and their carers/family members. Individuals will be recruited from my initial PPI session, the Psoriasis Association and the well-established inflammatory skin disease PPI network led by my supervisors at the St John's Institute of Dermatology. This will ensure a diversity of views, backgrounds and experiences.

The group will meet biannually, with additional meetings around key decision points. Based on feedback, meetings will be virtual to minimise disruption to daily schedules and maximise inclusivity. To reduce the potential for digital exclusion, I will arrange practice online calls and provide phone-in options.

PPI collaborators will:

- Provide input on variable selection for model development in Phase 2.
- Inform recruitment strategies for Phase 3, especially for underserved groups.
- Help design a topic guide and co-facilitate the focus groups in Phase 3.
- Assist with interpreting research findings and opine on their relevance to people with psoriasis and the public.

Additionally, they will review communications on research findings intended for patients and the public. Guided by the NIHR equality, diversity and inclusion toolkit, these will be widely disseminated via charities, established European patient networks and partners, newsletters and social media. I will also leverage the annual national PPI webinars co-organised by the Psoriasis Association and St John's DermAcademy (>100 attendees per event) to share my results. Where appropriate, I will invite PPI collaborators to co-author peer-reviewed publications and to co-present at events.

PPI collaborators will be invited to attend PPI training sessions run by Guy's and St Thomas' NHS Trust and by the NIHR Imperial College and Partners Research Support Service Hub. For those involved in Phase 3, I will provide additional training on conducting focus groups and interpreting qualitative data.

I have budgeted for PPI, including participant remuneration and expenses, in line with NIHR guidance[57]. GRIPP2 guidelines will be used to report on PPI activities[61].

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable.

8. Training & Development and Research Support

Proposed training and development programme

I have designed the following bespoke training and development plan with the aid of my supervisors. This will ensure I acquire the skills necessary to successfully deliver my proposal, and support my development as an academic dermatologist.

Systematic Reviews

I have experience with data extraction and risk of bias assessments through co-authoring two Cochrane systematic reviews. Additionally, I completed a 4-day **Cochrane Review Author Training Course** in 2018 and a 5-day **Systematic Reviews** course (Oxford University) in 2020.

I will build upon these foundations by attending a 4-day **Systematic Reviews and Meta-analysis of Health Research** course (London School of Hygiene & Tropical Medicine) in my first year, which will provide a refresher of systematic review methodology. This will be followed by a 3-day **Introduction to Meta-analysis** course (UCL), which will provide a practical toolkit for meta-analysis and heterogeneity assessment using R.

In addition to formal courses, I will also be supported via our group's collaboration with the Cochrane Skin Group.

Statistics and Computer Programming

I have developed a strong practical grounding in statistics over the past 6 years, with experience in summarising data using descriptive statistics and conducting survival analyses (see research background). I am also familiar with the RStudio environment and have significant prior experience in using R for data wrangling, visualisation and statistical tests. This has prepared me well for this fellowship, as R will be the main programming language used for data analysis.

I completed a 5-day **Statistical Computing with R and Stata** course in 2023 (Oxford University), which provided an introduction to statistical principles relevant to this project, including dealing with missing data, power calculations, regression modelling, and cross validation. While this gave me a basic understanding of different statistical techniques, I would benefit from further formal training during this fellowship.

I will attend a 5-day course on **Practical Statistics for Epidemiology using R** (Oxford University) in my first year to gain experience in interpreting and presenting statistical results, and learn techniques to address bias and confounding. I will also complete a 2-day **Regressions with R** course (UCL) to learn how to fit different regression models, including logistic regression which is central to my analyses. To prepare for the predictive modelling component of my proposal, I will attend a 3-day **Statistical Methods for Risk Prediction and Prognostic Models** course (Birmingham University). This will improve my knowledge of statistical methods for developing and validating predictive models. Additionally, I will receive dedicated hands-on training from Dr Niamh Dooley, a statistician at KCL, and Dr Sam Norton, who will supervise the statistical components of my project.

Genetic Epidemiology

In Phase 2, I will explore whether the predictive performance of clinical models can be meaningfully improved by incorporating genetic data. To better understand different genetic data types and the construction and use of polygenic risk scores, I will attend a 5-day **Genetic Epidemiology** course (Bristol University).

I will also benefit from a collaboration with Professor Michael Simpson at the Department of Genetics in KCL, who has significant expertise in this area.

Qualitative Research Methods

During my third year, I will conduct focus groups to collate stakeholder perspectives on the models developed. In preparation, I will complete two courses in my second year: a 1-day **Introduction to doing Qualitative Interviews** course followed by a 5-day **Introduction to Qualitative Research Methods** course (both Oxford University). These will yield vital knowledge on how to successfully conduct focus groups, and give me practice with coding and thematic analysis. I will build on this by completing a part-time **NVivo Core Skills Online Course** (NVivo Academy).

Dashboard Development

I plan to present the risk stratification tool as an interactive R Shiny dashboard. In my third year, I will complete the **Publishing Visualisations in R with Shiny and flexdashboard** course (Johns Hopkins University via Coursera). I will build on this by attending a 2-day **Interactive Dashboards & Web Apps using R & Shiny** course (Royal Statistical Society). I will also leverage expertise available within our group, where a project is currently underway to build an R Shiny dashboard to present pharmacokinetic/pharmacodynamic models of biologics.

PPI/E

Prior to starting the fellowship, I will have completed **Public Involvement Training** organised by the Patient Experience Research Centre at Imperial College London. During the first year of my fellowship, I will attend the **Online PPI training for researchers** run by the NIHR Cambridge Biomedical Research Centre PPI team. I am particularly interested in the sessions on building and maintaining a PPI group, using PPI to help communicate research findings, evaluating PPI and planning inclusive PPI. In addition, I will build on my skills in public communication by attending a 1-day **Public Engagement** course (Science Museum).

Our group has well-established links with the Psoriasis Association and a local inflammatory skin disease PPI network. I will benefit from this extensive expertise in involving patient partners in research.

Leadership

In my first year, I will complete a 6-day **Emerging Clinical Leaders** course (The King's Fund). This will provide valuable leadership training which will maximise the success of this project. In my third year, I will attend a 3-day **Future Leaders Academy** (European Society for Dermatological Research), focusing on delegation, conflict resolution, and influencing strategic change. This presents a unique opportunity to connect with future leaders in dermatology and receive guidance from experienced mentors, greatly benefitting my postdoctoral career.

Research Environment

My supervisors have considerable experience supporting PhD students to completion and will provide expertise on key methodological elements of this proposal. I will be hosted by the St John's Institute of Dermatology, a world-renowned centre of excellence in skin research with an outstanding track record of supporting clinical academic PhD fellows. The KCL Doctoral Development Programme offers a comprehensive suite of training opportunities, including courses on effective communication, teaching methods, and career management. Furthermore, the frequent seminars and institute-wide events foster a vibrant environment conducive to collaboration and creativity. Collectively, these factors make KCL an ideal home for this project and a nurturing space for my growth as an

academic dermatologist.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Dr Satveer Mahil

Institution	St John's Institute of Dermatology, King's College London (KCL)

Position	Consultant Dermatologist and Reader

Description of support provided
Dr Satveer Mahil is a Consultant Dermatologist and Reader at KCL. Her career to date has followed the NIHR Integrated Academic Training pathway, completing an NIHR Academic Clinical Fellowship which led to an MRC-funded PhD and subsequently an NIHR Clinical Lectureship. Post-PhD, she was awarded an MRC-NIHR Clinical Academic Research Partnership Award and she currently holds an NIHR Advanced Fellowship. Dr Mahil's translational research focuses on using large-scale clinical data and molecular profiling technologies to personalise management strategies for inflammatory skin diseases including psoriasis. She has a strong interest in optimising the management of individuals with psoriasis who have sustained disease control. She is currently leading an NIHR-funded multicentre feasibility randomised controlled trial comparing patient-led 'as-needed' biologic treatment, therapeutic drug monitoring-guided biologic treatment, and biologic continuation (standard care) in this cohort. She has extensive experience across multiple methods which will be used in this fellowship, including systematic reviews, observational dataset analysis and qualitative research. She is currently supervising 3 PhD students (all ongoing), including 1 NIHR DRF. She is co-director of the COVID-19 psoriasis real-world registries PsoProtect and PsoProtect*Me*, and is a member of British Association of Dermatologists and European Dermatology Forum psoriasis guideline committees. She is director of St John's DermAcademy, a national dermatology education platform. I have established an excellent working relationship with Dr Mahil throughout the development of this fellowship proposal and she will be my primary supervisor. We will meet weekly to discuss progress, address unforeseen challenges/barriers, and ensure my training needs are met.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Dr Sam Norton

Institution	King's College London (KCL)

Position	Reader in Research Methods and Statistics

Description of support provided
Dr Sam Norton is a Reader in Research Methods and Statistics at KCL. He has significant expertise in meta-analysis, longitudinal data analysis, and working with large-scale observational cohort studies of chronic disease. He is a KCL Clinical Trials Unit affiliated senior statistician and the capacity development lead for the NIHR Research Support Service delivered by KCL and its partners. He was previously the statistical lead for the NIHR Research Design Service London.

Dr Norton has considerable experience in the development of risk prediction tools and is currently supervising an NIHR DRF developing an infection prediction tool for people with immune mediated inflammatory diseases treated with targeted therapies. He has close collaborations with Dr Mahil, Professor Smith and Professor Coates, with a strong track record of producing high-impact research outputs.

I have worked closely with Dr Norton to design the data analysis plan for the quantitative aspects of this research proposal. Throughout the fellowship, he will directly supervise my statistical training to ensure I gain the necessary skills to perform the analyses required in each phase of the project proposal.

Dr Norton has supervised 21 doctoral students, of which 12 have completed and 9 are ongoing. He has considerable experience supporting NIHR fellowships and is a supervisor or collaborator for 2 current NIHR DRFs, 1 NIHR ACF, and 1 completed NIHR DRF. We will aim to meet fortnightly during the fellowship to discuss progress and ensure my training needs are met.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Professor Catherine Smith

Institution	St John's Institute of Dermatology, King's College London (KCL)

Position	Consultant Dermatologist, NIHR Senior Investigator and Professor of Dermatology and Therapeutics

Description of support provided

Professor Catherine Smith is a Consultant Dermatologist, NIHR Senior Investigator and Professor of Dermatology and Therapeutics at KCL. She directs the Skin Therapy Research unit and is a world-renowned psoriasis expert, with experience ranging from biomarker discovery to interventional studies. She sits on the Steering Committees of the key datasets for this fellowship: the British Association of Dermatologists Biologics and Immunomodulators Register (BADBIR), the Biomarkers and Stratification To Optimise outcomes in Psoriasis study (BSTOP), and the HIPPOCRATES Prospective Observational Study (HPOS). Additionally, she is involved in major national and international psoriasis collaborations, including the COVID-19 psoriasis registries, the EU BIOMAP consortium, and is a board member of the International Psoriasis Council. She also has strong links to the Psoriasis Association, chaired the NICE psoriasis guideline committee and leads the British Association of Dermatologists biologics guideline committee. She is the Integrated Academic Training Programme Lead at St John's Institute of Dermatology, as well as an NIHR mentor.

Professor Smith has supervised 7 PhD students to completion, and currently supervises 6 PhD students (including 1 NIHR DRF). She will ensure I have access to the relevant datasets and support throughout this fellowship. Her expert guidance on evidence synthesis, observational dataset analysis and PPI activities will also be invaluable. Through her extensive collaborations and networks, I will be exposed to relevant career development opportunities both during and after this fellowship. She will also facilitate the dissemination of my research findings to an international audience. We will aim to meet monthly to discuss progress.

RESEARCH SUPPORT / SUPERVISOR 4

Name	Professor Laura Coates

Institution	University of Oxford

Position	Associate Professor and Honorary Consultant Rheumatologist

Description of support provided
Professor Laura Coates is an Associate Professor and Honorary Consultant Rheumatologist at the University of Oxford. She has supported and mentored me throughout my clinical and academic training, starting from my time as an Academic Foundation Programme doctor. We have established an excellent working relationship as a result. Professor Coates is an international expert in psoriatic arthritis, with extensive experience in early diagnosis, outcome measure development, clinical trial design and systematic review. She is a Steering Committee Member of the Group for Research and Assessment of Psoriasis and Psoriatic arthritis (GRAPPA) and the British Psoriatic Arthritis Consortium (BritPACT). She is also a member of the British Association of Dermatologists psoriasis guideline committee. Professor Coates has an excellent track record in engaging PPI collaborators in her research, and will provide guidance on PPI activities throughout this fellowship. In addition, she is the Chief Investigator for HPOS, which provides the validation dataset for my project. Professor Coates currently supervises 5 PhD students (including 1 NIHR DRF), 3 pre-doctoral fellows and 1 international post-doctoral fellow. Her supervisees include rheumatologists, a dermatologist, physiotherapist, lawyer and statistician. She also mentors two PIs nationally. During this fellowship, she will provide essential guidance in examining the subset of individuals with concurrent psoriatic arthritis. I will leverage her broad network of connections throughout rheumatology to ensure widespread dissemination of my findings to allied medical disciplines. I will aim to meet Professor Coates monthly, with increased contact depending on the phase of the project and my training needs.

Collaborations
Professor Emilie Sbidian and Professor Laurence Le Cleach Professor Emilie Sbidian is a Professor of Dermatology and Therapeutics in the Department of Dermatology at Henri Mondor Hospital, University Paris Est Creteil in France. She is a core member of the Cochrane Skin Group and has led on the living Cochrane systematic review and network meta-analysis which examines systemic pharmacological treatments for chronic plaque psoriasis. Professor Laurence Le Cleach is an Associate Professor of Dermatology and Therapeutics at Henri Mondor Hospital, University Paris Est Creteil in France. She is the coordinating editor of the Cochrane Skin Group. Professor Sbidian and Professor Le Cleach are existing collaborators with our group. Through this, I will be able to gain access to a wealth of resources which will contribute to answering my review question as outlined in *Section 5.1 – Design* of the detailed research plan. **Professor Michael Simpson**

Professor Michael Simpson is the Head of the Department of Medical and Molecular Genetics at King's College London. His research focuses on understanding the molecular and cellular mechanisms through which genetic variation influences susceptibility to skin disease and treatment response.

Professor Simpson is a close collaborator with our group in ongoing work aimed at examining the genetic determinants of treatment response. He will be able to provide support with the analysis outlined in *Section 5.2 – Data analysis*, which explores whether the predictive performance of clinical models can be meaningfully improved by incorporating genetic data.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

R.e. Dr Weiyu Ye – Head of Department Supporting Statement

My name is John McGrath, and I am the Academic Head of St John's Institute of Dermatology within King's College London (Guy's Hospital campus). It is a pleasure to be asked to write a supporting statement for Weiyu Ye's NIHR Doctoral Fellowship Application.

I confirm that St John's Institute of Dermatology, part of King's College London (KCL), supports this application and that, if funded, the research and training will be supported and administered by KCL.

I confirm that Weiyu Ye will undertake this work and is eligible, for the duration of the award, to live and work in the UK.

Dr Ye wishes to develop a clinical academic career and I confirm that I am aware of the principles and obligations published by NIHR to support her clinical academic training and I confirm that KCL is committed to these principles and obligations.

The St John's Institute of Dermatology (School of Basic and Medical Biosciences, KCL) is an internationally renowned centre for the delivery of innovative clinical care for patients with skin diseases, cutting-edge education, training, and research. A recent independent RAND assessment ranked St John's as the leading clinical and research institution for dermatology in the UK. St John's occupies 1,200 square metres of state-of-the-art laboratory space (Guy's Hospital campus) and is fully integrated into the Faculty of Life Sciences & Medicine's School of Basic & Medical Biosciences. An active programme of graduate education is pursued including research training (PhD and MD) and a world-renowned taught course (MSc) in Clinical Dermatology. The Institute excels in research training, with a >95% success record of its Integrated Academic Training Pathway Fellows obtaining external funding from MRC/NIHR. In addition, St John's Institute of Dermatology plays a leading role in the training of dermatologists, both in the UK and around the world. Research at St John's targets 4 key areas: inflammatory and autoimmune skin disorders (the key area of this research proposal), cutaneous oncology, genetic skin disorders, and skin-environment interactions. Our research is supported by grants from a range of sources including NIHR, MRC, The Wellcome Trust, EU, US Foundations, British Skin Foundation, national charities, and commercial sources. Our department research income averages >£3m per year with a School average research grant income of >£25m per year. The latest submission to the REF confirmed Clinical Medicine was independently rated to have 90% of its research as world leading or internationally excellent (3 or 4*) and 84 % of impact as world leading (4*). King's College London is part of King's Health Partners, one of the UK's five Academic Health Sciences Centres, where world-class research, teaching and clinical practice are brought together for the benefit of patients. The three NHS partners; Guy's and St Thomas', King's College Hospital (KCH) and South London and Maudsley NHS Foundation Trusts, are among the highest rated Trusts in the country and compete on the world stage and attract the best clinical academics to pioneer world-leading advances in health and well-being. Dr Weiyu Ye will have full support from the extensive infrastructure within the Department, School, KCL, and King's Health Partners. We will ensure the equipment and facilities successfully allow Weiyu to pursue her research. We will also provide extensive personal support and mentorship to allow Weiyu to fully develop her research ideas, and also to embed her career as a future leading clinical academic in dermatology. As a host organization, KCL is committed to creating and maintaining an inclusive and supportive research culture for all, championing equality, diversity and inclusion, and research integrity. KCL has been a member of the Athena SWAN Charter since 2007 and in 2021 it received a Silver Award. In 2018, the Faculty of Life Sciences & Medicine was recognised with a Faculty Athena

SWAN Silver award as recognition of our ongoing commitment to achieving gender equality and an inclusive environment.

Dr Ye's project aims to develop a risk stratification tool to predict the likelihood of sustaining psoriasis control on systemic therapy. This has the potential to facilitate more personalised long-term management for individuals who have achieved psoriasis control, thereby improving patient outcomes. The project has 3 main workstreams:

1. Systematic review to evaluate published evidence on the proportion of individuals sustaining psoriasis control over time on systemic therapy, and the factors associated with this.
2. Model the likelihood of sustaining disease control on systemic therapy using large-scale real-world datasets.
3. Optimise the predictive models for use as a risk stratification tool, with patient and clinician input.

This work is integrated with the clinical academic activities of two Faculty members within the department, Professor Catherine Smith and Dr Satveer Mahil, who will provide direct supervision, training and mentoring.

In short, I am delighted to offer this Head of Department letter of support, both for Weiyu as an excellent individual embarking on her clinical-academic training and for the impressive nature of the scientific proposal I have discussed with her and her supervisors. We offer full institutional support for this application.

Please do not hesitate to contact me if any further information is required.

Yours sincerely,

Professor John McGrath MD FRCP FMedSci

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Weiyu Ye

Employing organisation	King's College London

Grade	ST5, Dermatology Specialist Registrar

Starting point on scale	ST5

Increment Date	01/04/2025

Justification of Costs
Full-time salary for the lead applicant for the duration of the 3-year Doctoral Fellowship, based on nationally agreed NHS pay scales.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	55329
Geographical Weighting (£)	0
Other allowances (£)	287
Superannuation (£)	8267
Nat. Insurance (£)	7283
Current Annual Costs (£)	**71166**

Staff Type: (Lead Applicant)-Dr Weiyu Ye
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: Full-time salary for the lead applicant for the duration of the 3-year Doctoral Fellowship, based on nationally agreed NHS pay scales.

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	59241	63152	63152	185545
National Insurance	7218	7758	7758	22734

Superannuation	8903	9471	9471	27845
Geographical weighting / Other allowances (if applicable)	2469	2489	2489	7447

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	194856	0	0	194856
Total NHS	0	0	0	0
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference Attendance Costs
Justification of Cost: Conference attendance costs including the Annual Meeting of the British Association of Dermatologists (BAD), the British Society for Investigative Dermatology (BSID) Annual Meeting, the European Society for Dermatological Research (ESDR) Annual Meeting, and the American Academy of Dermatology (AAD) Annual Meeting.

	Year 1	Year 2	Year 3	Total Cost (£)
	200	1400	1400	3000

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	200	1400	1400	3000

Equipment

Type of cost: HEI
VAT relief to be applied: False
Description: Laptop
Price excluding VAT £: 2500
Expected life time: 36
Justification of Cost: As my project involves working with very large datasets (including genetic data) and has a heavy analytical and model-fitting element, I will require a high-performance laptop with a powerful CPU, plentiful RAM and at least 1TB of SSD storage.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	2500	0	0	2500

Type of cost: HEI
VAT relief to be applied: False
Description: Monitor
Price excluding VAT £: 500
Expected life time: 36
Justification of Cost: I have requested funds for a secondary monitor as this will significantly improve my productivity, especially considering the heavy programming component of my project.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	500	0	0	500

Type of cost: HEI
VAT relief to be applied: False
Description: Laptop accessories
Price excluding VAT £: 100
Expected life time: 36
Justification of Cost: I have also requested funds for appropriate laptop accessories, including a keyboard, mouse, and case.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	100	0	0	100

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	3100	0	0	3100

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	0	3000	2480
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables
No costs have been added

Patient and Public Involvement

Description: Patient and Lay Advisory Group Meetings
Type of cost: HEI
Please describe fully: Patient and Lay Advisory Group Meetings
Justification of Cost: Patient and Lay Advisory Group Meetings - I have budgeted for 6 participants to meet virtually for up to 2 hours at 6-monthly intervals, with 2 additional meetings at key decision points (£50/meeting + £5 online connection charge, for 8 times during the fellowship, in line with NIHR guidance https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	880	880	880	2640

Description: Patient and Lay Advisory Group co-facilitation of qualitative focus groups
Type of cost: HEI
Please describe fully: Patient and Lay Advisory Group co-facilitation of qualitative focus groups
Justification of Cost: In Phase 3 of my project, a trained member of the Patient and Lay Advisory Group will co-facilitate each focus group (6 sessions up to 2 hours at £50 each + £5 online connection charge, in line with NIHR guidance https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	330	0	330

Description: Qualitative data analysis training for PPI collaborators
Type of cost: HEI
Please describe fully: Qualitative data analysis training for PPI collaborators
Justification of Cost: 2 PPI collaborators will be involved in analysing the focus group discussions. Prior to this,

they will attend a half day online training session (£75 + £5 connection charge, in line with NIHR guidance https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	160	0	160

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	880	1370	880	3130

Other Direct Costs

Description: Payment for qualitative focus groups
Type of cost: HEI
Please describe fully: Payment for qualitative focus groups
Justification of Cost: Payment for qualitative focus groups (48 individuals for up to 2 hours at £50 each + £5 connection charge, in line with NIHR guidance https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392)

	Year 1	Year 2	Year 3	Total Cost (£)
	0	2640	0	2640

Description: Payment for beta testing
Type of cost: HEI
Please describe fully: Payment for beta testing
Justification of Cost: Payment for beta testing (20 individuals for up to 2 hours at £50 each + £5 connection charge, in line with NIHR guidance https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392). I have budgeted for 2 beta testing sessions during the fellowship.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	2200	2200

Description: Transcription costs for focus groups
Type of cost: HEI
Please describe fully: Transcription costs for focus groups
Justification of Cost: Funds to cover transcription costs for focus groups (6 focus groups at 90 mins each, at £1.85/minute).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	999	0	999

Description: Data storage costs
Type of cost: HEI
Please describe fully: Data storage costs
Justification of Cost: As large scale datasets will be used during this project, funds have been requested to cover data storage, back up and security.

	Year 1	Year 2	Year 3	Total Cost (£)
	1700	1700	1600	5000

Description: Dashboard development and hosting costs
Type of cost: HEI
Please describe fully: Dashboard development and hosting costs
Justification of Cost: Funds have been requested to cover the development and hosting of the online dashboard.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	30000	30000

Description: BADBIR data management costs
Type of cost: HEI
Please describe fully: BADBIR data management costs

Justification of Cost: Funds have been requested to cover the data management costs for BADBIR (http://www.badbir.org/), which is the main dataset for exploratory analysis and internal validation in this project.

	Year 1	Year 2	Year 3	Total Cost (£)
	8600	0	0	8600

Description: BSTOP data management costs
Type of cost: HEI
Please describe fully: BSTOP data management costs
Justification of Cost: Funds have been requested to cover the data management costs for BSTOP (https://www.kcl.ac.uk/research/bstop), which contains genetic data required for the predictive modelling component of this project.

	Year 1	Year 2	Year 3	Total Cost (£)
	5000	0	0	5000

Description: HPOS data management costs
Type of cost: HEI
Please describe fully: HPOS data management costs
Justification of Cost: Funds have been requested to cover the data management costs for HPOS (https://hpos.study/), which is a large-scale European dataset that I will use to externally validate BADBIR findings.

	Year 1	Year 2	Year 3	Total Cost (£)
	8000	0	0	8000

Description: Covidence subscription
Type of cost: HEI
Please describe fully: Covidence subscription
Justification of Cost: Covidence will be used to manage the systematic review component of my proposal. I will purchase a subscription for the first two years, at £250/year.

	Year 1	Year 2	Year 3	Total Cost (£)
	250	250	0	500

Description: Part time statistician costs 0.1 WTE
Type of cost: HEI
Please describe fully: Part time statistician costs 0.1 WTE
Justification of Cost: Funding is sought for a 0.1 WTE statistician (Dr Niamh Dooley) in Years 2 and 3 to provide statistical oversight and guidance for the predictive modelling component of this project. I will be carrying out the statistical analysis myself and have built in a robust training programme to ensure I will be well-equipped to deliver on the analysis. The statistician will provide guidance during the analysis process. Dr Niamh Dooley is a Research Associate Statistician at KCL and her costings have been provided by her department.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	6248	6420	12668

Description: Guy's and St Thomas' NHS Foundation Trust Archiving Fee
Type of cost: HEI
Please describe fully: Guy's and St Thomas' NHS Foundation Trust Archiving Fee
Justification of Cost: An archiving fee of £565 has been quoted by the Guy's and St Thomas' NHS Foundation Trust as part of their role in faciltiating participant recruitment for the qualitative work in Phase 3

	Year 1	Year 2	Year 3	Total Cost (£)
	0	565	0	565

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	23550	12402	40220	76172

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD Tuition Fees
Justification of Cost: Funds to cover KCL PhD fees

	Year 1	Year 2	Year 3	Total Cost (£)
	6936	7352	7720	22008

Type of cost: HEI
Description: Short Courses
Item Description: Systematic Reviews and Meta-analysis of Health Research, London School of Hygiene and Tropical Medicine
Justification of Cost: In preparation for the systematic review component of my project, I will attend this 4-day course organised by the London School of Hygiene & Tropical Medicine (https://www.lshtm.ac.uk/study/courses/short-courses/systematic-reviews-research). Funds have been requested to cover course fees

	Year 1	Year 2	Year 3	Total Cost (£)
	950	0	0	950

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to Meta-analysis, University College London
Justification of Cost: 3-day course organised by University College London, which will provide a practical toolkit for meta-analysis and heterogeneity assessment using R (https://www.ucl.ac.uk/short-courses/search-courses/introduction-meta-analysis). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	225	0	0	225

Type of cost: HEI
Description: Short Courses
Item Description: Practical Statistics for Epidemiology using R, University of Oxford
Justification of Cost: I will attend this 5-day course organised by the University of Oxford in my first year, to gain experience in processing and visualising epidemiological data using R, interpreting and presenting statistical results, and learn techniques to address bias and confounding (https://www.ndph.ox.ac.uk/study-with-us/practical-statistics-for-epidemiology-using-r). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	500	0	0	500

Type of cost: HEI
Description: Short Courses
Item Description: Regressions with R, University College London
Justification of Cost: I will attend this 2-day course to learn how to fit different regression models in R, including logistic regression which is central to my analyses (https://www.ucl.ac.uk/short-courses/search-courses/regressions-r-overview). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	350	0	0	350

Type of cost: HEI
Description: Short Courses
Item Description: Statistical methods for risk prediction and prognostic models , University of Birmingham
Justification of Cost: I will attend this 3-day course to improve my knowledge of statistical methods for developing and validating predictive models (https://www.birmingham.ac.uk/postgraduate/courses/cpd/med/statistical-methods-for-risk-prediction-and-prognostic-models.aspx#CourseOutcomeTab). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	499	0	0	499

Type of cost: HEI
Description: Short Courses
Item Description: Genetic Epidemiology, University of Bristol
Justification of Cost: I will attend this 5-day course to improve my understanding of different genetic data types and the construction and use of polygenic risk scores, which will be useful for the predictive modelling component of my proposal (https://www.bristol.ac.uk/medical-school/study/short-courses/courses/genetic-epidemiology/). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	1100	0	0	1100

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to doing Qualitative Interviews, University of Oxford
Justification of Cost: I will complete this 1-day course to learn about the practical and theoretical elements of qualitative research interviews, including how to produce a topic guide (https://www.phc.ox.ac.uk/study/short-courses-in-qualitative-research-methods/introduction-to-qualitative-interviewing). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	480	0	480

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to Qualitative Research Methods, University of Oxford
Justification of Cost: In preparation for the qualitative component of my project, I will attend this highly regarded 5-day course on qualitative research methods (https://www.phc.ox.ac.uk/study/short-courses-in-qualitative-research-methods/introduction-to-qualitative-research-methods). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	1750	0	1750

Type of cost: HEI
Description: Short Courses
Item Description: Public Engagement, Science Museum
Justification of Cost: I will attend a 1-day course organised by the Science Museum to build skills in public communication (https://learning.sciencemuseumgroup.org.uk/academy/academy-training/public-engagement/). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	450	0	450

Type of cost: HEI
Description: Short Courses
Item Description: NVivo Core Skills Online Course, NVivo Academy
Justification of Cost: I will complete this part-time online course organised by the NVivo Academy to gain familiarity with using the NVivo software, which will be used to analyse the qualitative data during my project (https://lumivero.com/product/nvivo-core-skills-online-course/). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	50	0	50

Type of cost: HEI
Description: Short Courses
Item Description: Interactive Dashboards & Web Apps using R & Shiny, Royal Statistical Society
Justification of Cost: I will complete this 2-day course organised by the Royal Statistical Society to learn about dashboard development using R Shiny (https://rss.org.uk/training-events/training/public-courses/software-training/interactive-dashboards-web-apps-using-r-shiny/interactive-dashboards-web-apps-using-r-shiny-(2)/). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)

	0	0	927	927

Type of cost: HEI
Description: Short Courses
Item Description: Emerging Clinical Leaders, The King's Fund
Justification of Cost: I will attend this 6-day course organised by The King's Fund, which will provide invaluable leadership training (https://www.kingsfund.org.uk/courses/emerging-leaders). Funds have been requested to cover course fees.

	Year 1	Year 2	Year 3	Total Cost (£)
	3114	0	0	3114

Type of cost: HEI
Description: Short Courses
Item Description: Future Leaders Academy, European Society for Dermatological Research
Justification of Cost: I will attend this 3-day course organised by the European Society for Dermatological Research in collaboration with the Manchester Business School, which focuses on delegation, conflict resolution and influencing strategic change (https://esdr.org/events/esdr-future-leaders-academy-2024/). In addition, this provides a unique opportunity to connect with future leaders in dermatology and receive guidance from experienced mentors, which will be greatly beneficial to my postdoctoral career. I plan to attend in the third year of my fellowship, and funds have been requested to cover European travel and accommodation for 3 nights (location to be determined).

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	800	800

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	13674	10082	9447	33203

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	98162	88795	110023	296980

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: King's College London
Justification of Cost: HEI estate costs have been calculated by the King's College London WorkTribe Costing System.

	Year 1	Year 2	Year 3	Total Cost (£)
	17731	19504	19504	56739

Type of cost: Other Indirect Costs
Name of Higher Education Institute: King's College London
Justification of Cost: Indirect cost rate. HEI indirect costs have been calculated by the King's College London WorkTribe Costing System.

	Year 1	Year 2	Year 3	Total Cost (£)
	69806	76787	76787	223380

	Year 1	Year 2	Year 3	Total Cost (£)

HEI Indirect costs Total	87537	96291	96291	280119

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	70030	77033	77033	224096

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	260777	224096	484873
Total Higher Education Institution Costs (100%)s	36203	0	36203
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	0	0	0
Total Research Costs Requested From Funder	296980	224096	521076

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**62264**	**66296**	**66296**	**194856**
Travel-subsistence-and-conference-fees	**200**	**1400**	**1400**	**3000**
Equipment	**2480**	**0**	**0**	**2480**
Consumables	**0**	**0**	**0**	**0**
Patient-and-Public-Involvement	**704**	**1096**	**704**	**2504**
Other-Direct-Costs	**18840**	**9921**	**32176**	**60937**
Training And Development	**13674**	**10082**	**9447**	**33203**
HEI-Indirect-costs	**70030**	**77033**	**77033**	**224096**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**168192**	**165828**	**187056**	**521076**

NHS Support Costs
No costs have been added

Have you discussed and agreed these costs with the Lead	Yes

Network?	

NHS Treatment Costs
No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
N/A

Usual Treatment Costs
No costs have been added

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	521076
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	521076

10. Uploads

(References - References.pdf) is included as an appendix within this file.

(Figures / Tables - Figure.pdf) is included as an appendix within this file.

(Research Timetable - Gantt Chart.pdf) is included as an appendix within this file.

(CTU Letter of Support - Cochrane Skin Group Letter of Support.pdf) is included as an appendix within this file.

(SoECAT Template - SoECAT WY_Approved_EF_7.12.2023.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Devolved Administrations

no Devolved Administrations Participant has been added

Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Dr
Forename(s)	Satveer
Surname	Mahil
Position	Consultant Dermatologist and Adjunct Senior Lecturer

Supervisor

Supervisor Participant

Title	Dr
Forename(s)	Laura
Surname	Coates
Position	Senior Lecturer

Supervisor Participant

Title	Dr
Forename(s)	Sam
Surname	Norton
Position	Lecturer in Research Methods & Statistics

Supervisor Participant

Title	Professor
Forename(s)	Catherine
Surname	Smith
Position	Professor of Dermatology & Therapeutics

Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Professor
Forename(s)	John

Surname	McGrath
Position	Mary Dunhill Chair in Cutaneous Medicine

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
Forename(s)	Ania
Surname	Szefer
Position	Research Funding Associate

12. Acknowledgement, Review and Submit

Conflict checks:
Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.

Agreement to Terms and Conditions:
I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the NIHR Research Governance Framework.

If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. References.pdf
2. Figure.pdf
3. Gantt Chart.pdf
4. Cochrane Skin Group Letter of Support.pdf
5. SoECAT WY_Approved_EF_7.12.2023.xlsx

Appendix #1: References.pdf

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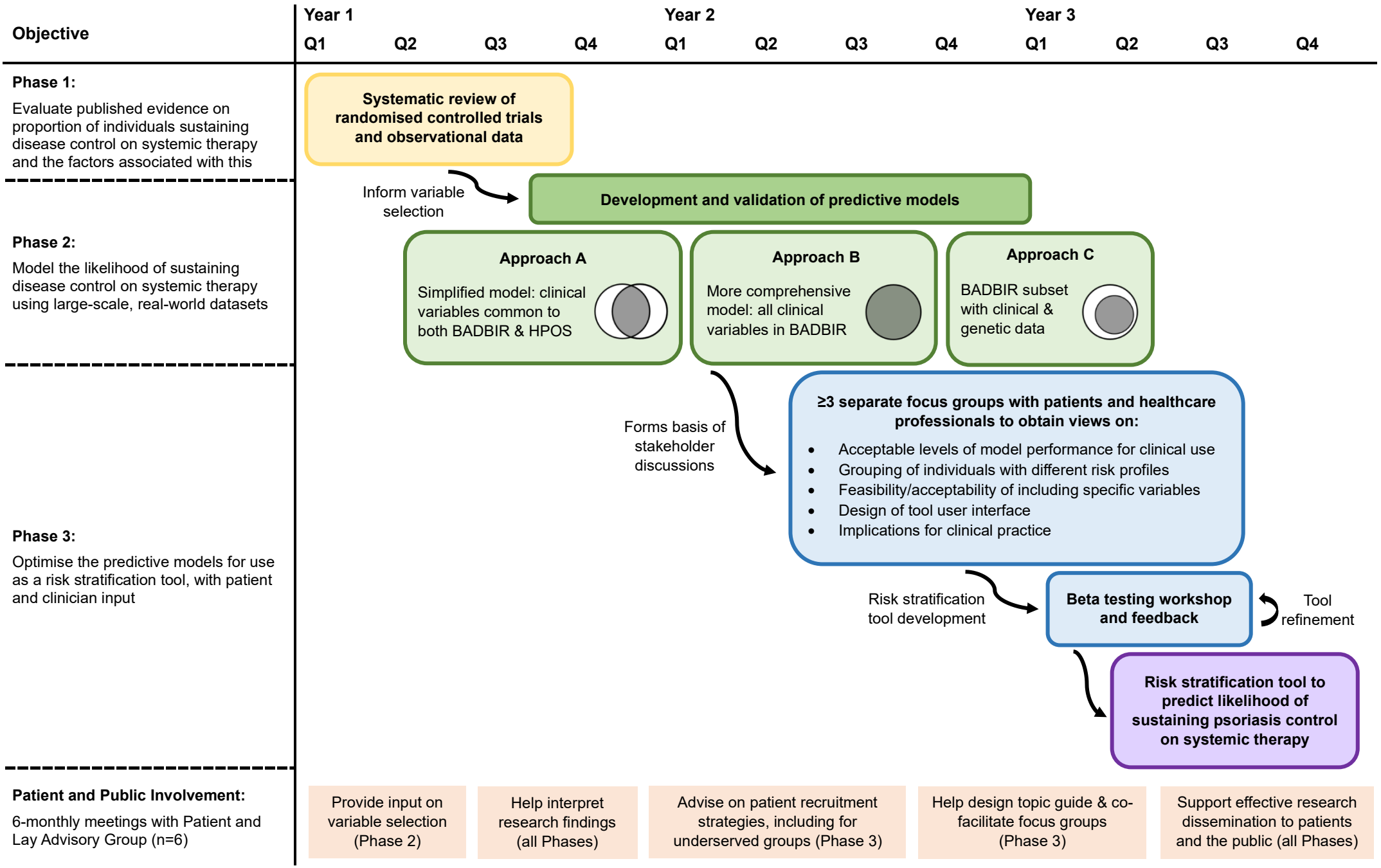
Appendix #2: Figure.pdf

Predicting long-term outcomes in people with psoriasis who achieve disease control on systemic therapy: a mixed methods study



Appendix #3: Gantt Chart.pdf

NIHR Doctoral Fellowship Research Timetable

		Months											
		0-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24	25-27	28-30	31-33	34-36
Project work													
Phase 1: Systematic review	Register protocol												
	Perform searches, screening, data extraction												
	Data analysis and quality assessment												
	Write up and submit publication												
Phase 2: Model development	Assess HPOS recruitment progress												
	Apply for PsoBest access (if required)												
	Data transfer and preparation												
	Data analysis and predictive model development												
	Model validation (internal/external)												
	Write up and submit publication												
Phase 3: Developing the risk stratification tool	Obtain ethical approval												
	Participant recruitment												
	Focus groups and qualitative data analysis												
	Tool development												
	Beta testing and tool refinement												
	Write up and submit publication												
	Write up and submit PhD thesis												
Training Courses													
Systematic Review & Meta-analysis courses													
Online PPI training for researchers													
Practical Statistics / Regressions with R													
Statistical Methods for Risk Prediction & Prognostic Models													
Genetic Epidemiology													
The King's Fund Leadership course													
Public Engagement course													
Introduction to Qualitative Interviews & Research Methods													
NVivo Core Skills Online Course													
R Shiny dashboard courses													
ESDR Future Leaders Academy													
Conferences													
Planned conference attendance (see key below)				BSID				BAD	ESDR			AAD	

Key: BSID (British Society for Investigative Dermatology Annual Meeting), BAD (Annual Meeting of the British Association of Dermatologists), ESDR (European Society for Dermatological Research Annual Meeting), AAD (American Academy of Dermatology Annual Meeting).

Appendix #4: Cochrane Skin Group Letter of Support.pdf

Cochrane Skin

Trusted evidence.
Informed decisions.
Better health.

12nd January 2024

Dear NIHR Doctoral Research Fellowship Selection Committee,

We are writing to confirm our support for Dr Weiyu Ye's application for an NIHR Doctoral Research Fellowship, titled '**Predicting long-term outcomes in people with psoriasis who achieve disease control on systemic therapy: a mixed methods study**'.

We would be delighted to support Dr Ye with the systematic review component of her research proposal, where she aims to evaluate published evidence on the proportion of individuals with psoriasis sustaining disease control over time on systemic therapy and the factors associated with this.

She will contribute as author to the following review: Serious adverse events associated with systemic treatments for psoriasis: a network meta-analysis of observational studies and randomised controlled trials (MS# COCHRANEEMD-2022-00477R3). As part of this, she will be involved in:

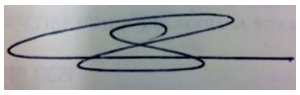
- Selection of observational studies
- Extraction of data on observational studies
- ROB2 assessment for RCTs and Robins assessment for observational studies

She will have access to:

- Completed data extraction for 167 RCTs including 12-month efficacy outcomes
- Data extraction from RCTs and observational trials (ongoing)

We look forward to working with her and are pleased to be collaborating on this exciting project.

Kind Regards,



Professor Emilie Sbidian
Member of Cochrane Skin

Professor Laurence Le Cleach
Coordinating Editor of Cochrane Skin

Appendix #5: SoECAT WY_Approved_EF_7.12.2023.xlsx